

Wireless IoT Network Protocols



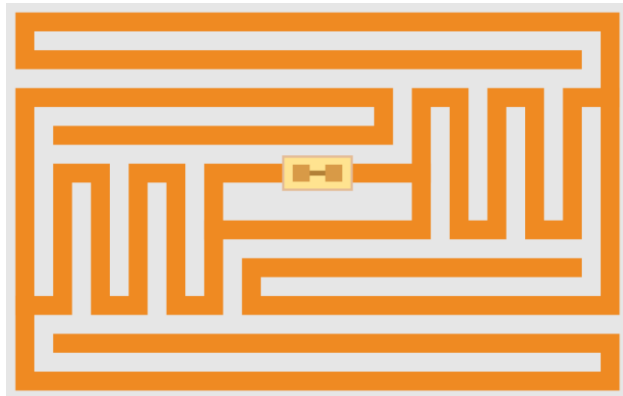
IoT

Networking

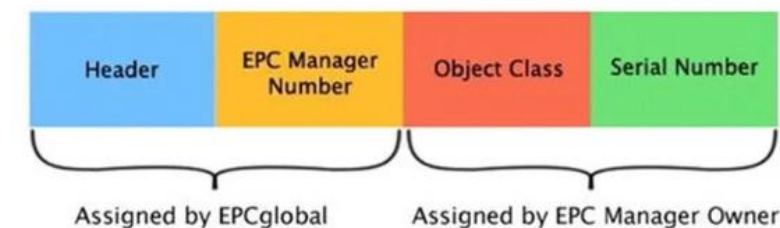


CoAP

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(Electronic Product Code)?



Functionality-based IoT Protocol Organization

- ✓ **Connectivity** (6LowPAN, RPL)
- ✓ **Identification** (EPC, uCode, IPv6, URIs)
- ✓ **Communication / Transport** (WiFi, Bluetooth, LPWAN)
- ✓ **Discovery** (Physical Web, mDNS, DNS-SD)
- ✓ **Data Protocols** (MQTT, CoAP, AMQP, Websocket, Node)
- ✓ **Device Management** (TR-069, OMA-DM)
- ✓ **Semantic** (JSON-LD, Web Thing Model)
- ✓ **Multi-layer Frameworks** (Alljoyn, IoTivity, Weave, Homekit)

Source: [Internet of Things Protocols \(Online\)](#)

Identification Protocols – EPC/

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Electronic Product Code (EPC): Unlocking the Power of Supply Chain Visibility

Revolutionize your supply chain with the Electronic Product Code (EPC) - a cutting-edge identification system that offers unparalleled visibility, efficiency, and security for physical goods.

EPC, short for Electronic Product Code, is a 96-bit code that is electronically encoded in RFID tags to identify various kinds of trade products including retail items such as apparel, shoes, food items

Electronic Product Code (EPC)



An Electronic Product Code (EPC) is a universal identifier that gives a unique identity to a specific physical object.



This identity is designed to be unique among all physical objects and all categories of physical objects in the world, for all time.



In most instances, EPCs are encoded on RFID tags which can be used to track all kinds of objects including: trade items, fixed assets, documents, or reusable transport items.

Beyond Barcodes: The Digital Passport of Products

Comprehensive Data

Unlike traditional barcodes, EPCs store dynamic information about a product's journey, from manufacturing to your doorstep.

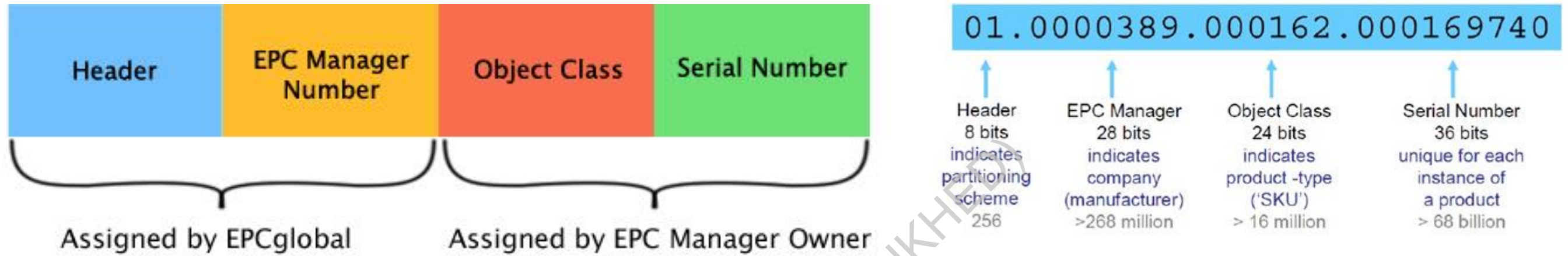
Standardized Infrastructure

The EPCglobal Network enables seamless data sharing across the supply chain, fostering unprecedented transparency.

Unique Identification

Each EPC acts as a digital fingerprint, allowing you to track and trace individual items with unparalleled precision.

Anatomy of an EPC



1 Header

Identifies the EPC version and data type.

2 EPC Manager Number

Indicates the organization responsible for the unique identifier.

3 Object Class

Categorizes the type of product.

4 Serial Number

Uniquely identifies the individual item within its object class.

The Backbone of EPC: RFID Technology



Wireless Communication

RFID tags communicate automatically with readers, enabling real-time data capture and tracking.

Diverse Functionality

Different types of RFID tags and readers offer varying capabilities to suit diverse supply chain needs.

Scalable Solutions

RFID infrastructure can be tailored to fit the unique requirements of your organization.

Transformative Impact

RFID is the key that unlocks the full potential of EPC, revolutionizing supply chain management.

Introduction

- ✓ RFID is an acronym for “radio-frequency identification”
- ✓ Data digitally encoded in RFID tags, which can be read by a reader.
- ✓ Somewhat similar to barcodes.
- ✓ Data read from tags are stored in a database by the reader.
- ✓ As compared to traditional barcodes and QR codes, RFID tag data can be read outside the line-of-sight.

Source: “[How does RFID work?](#)” AB&R (Online)

RFID Features

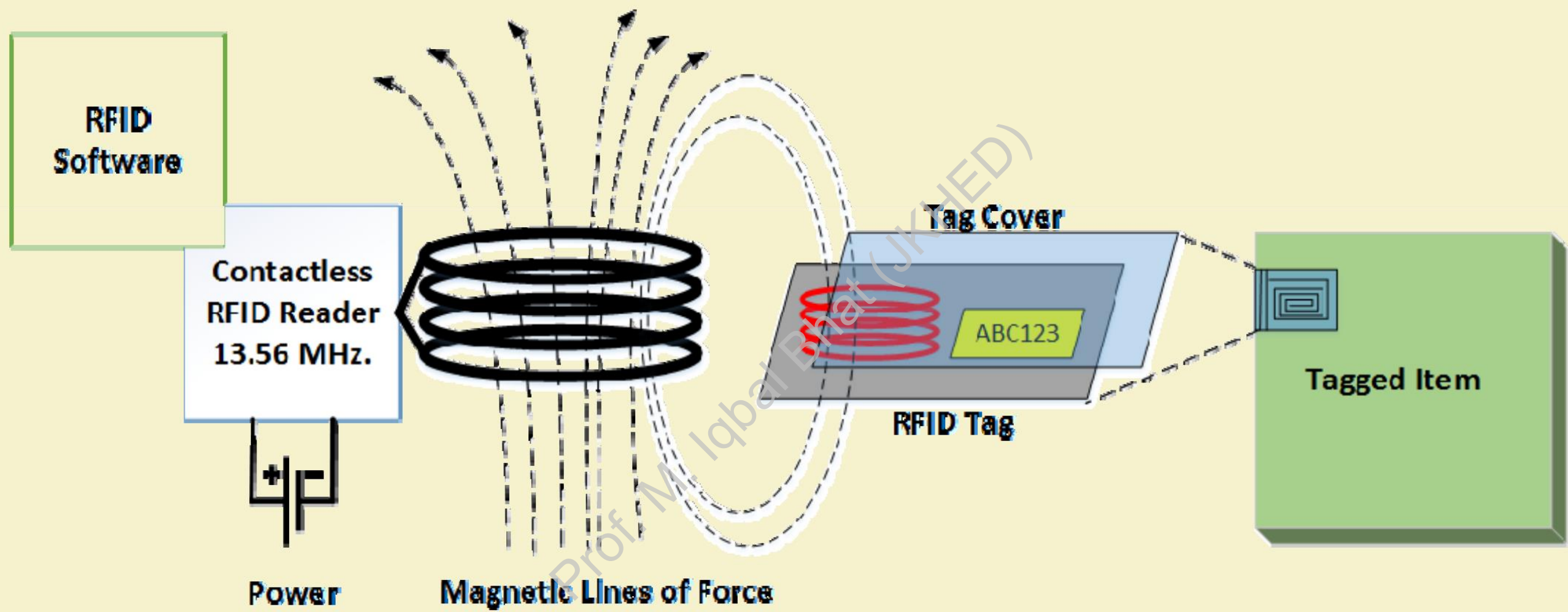
- ✓ RFID tag consists of an integrated circuit and an antenna.
- ✓ The tag is covered by a protective material which also acts as a shield against various environmental effects.
- ✓ Tags may be passive or active.
- ✓ Passive RFID tags are the most widely used.
- ✓ Passive tags have to be powered by a reader inductively before they can transmit information, whereas active tags have their own power supply.

Source: "[How does RFID work?](#)" AB&R (Online)

Working Principle

- ✓ Derived from Automatic Identification and Data Capture (AIDC) technology.
- ✓ AIDC performs object identification, object data collection and mapping of the collected data to computer systems with little or no human intervention.
- ✓ AIDC uses wired communication
- ✓ RFID uses radio waves to perform AIDC functions.
- ✓ The main components of an RFID system include an RFID tag or smart label, an RFID reader, and an antenna.

Source: "[How does RFID work?](#)" AB&R (Online)



Applications

- ✓ Inventory management
- ✓ Asset tracking
- ✓ Personnel tracking
- ✓ Controlling access to restricted areas
- ✓ ID badging
- ✓ Supply chain management
- ✓ Counterfeit prevention (e.g. in the pharmaceutical industry)

Source: "[How does RFID work?](#)" AB&R (Online)

Benefits of Embracing EPC/RFID



Enhanced Visibility

Gain real-time insights into inventory, product location, and supply chain movements.



Improved Efficiency

Automate data capture, streamline processes, and eliminate manual errors.



Increased Security

Protect against counterfeiting and ensure product authenticity.



Better Customer Experience

Offer faster delivery, accurate product information, and improved service.

EPC in Action: Transforming Industries

1

Retail

Optimize inventory management, streamline point-of-sale, and enhance customer engagement.

2

Manufacturing

Track production progress, ensure quality control, and manage assets effectively.

3

Logistics & Supply Chain

Streamline shipment tracking, optimize warehouse and yard management, and improve delivery efficiency.

4

Healthcare

Track pharmaceuticals, manage medical equipment, and ensure patient safety.

The Evolving Landscape of EPC

Wider Adoption

EPC/RFID technology is expected to be adopted across even more industries, driving further innovation and standardization.

Technological Integration

The integration of EPC with emerging technologies like blockchain and IoT will enhance transparency, security, and data-driven decision-making.

Evolving Standards

New and improved standards and protocols will address emerging needs and challenges, ensuring the continued growth and relevance of EPC.

Data Protocols

 **MQTT**

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Message Queue Telemetry Transport

Introduction

- ✓ **Message Queue Telemetry Transport.**
- ✓ ISO standard (ISO/IEC PRF 20922).
- ✓ It is a publish-subscribe-based lightweight messaging protocol for use in conjunction with the TCP/IP protocol.
- ✓ MQTT was introduced by IBM in 1999 and standardized by OASIS in 2013.
- ✓ Designed to provide connectivity (mostly embedded) between applications and middle-wares on one side and networks and communications on the other side.

Source: [“MQTT”, Wikipedia \(Online\)](#)

- ✓ A message broker controls the publish-subscribe messaging pattern.
- ✓ A topic to which a client is subscribed is updated in the form of messages and distributed by the message broker.
- ✓ Designed for:
 - Remote connections
 - Limited bandwidth
 - Small-code footprint

Source: ["MQTT", Wikipedia \(Online\)](#)

MQTT Components

Publishers

- Lightweight sensors

Subscribers

- Applications interested in sensor data

Brokers

- Connect publishers and subscribers
- Classify sensor data into topics

Source: ["MQTT", Wikipedia \(Online\)](#)

MQTT Methods

Connect

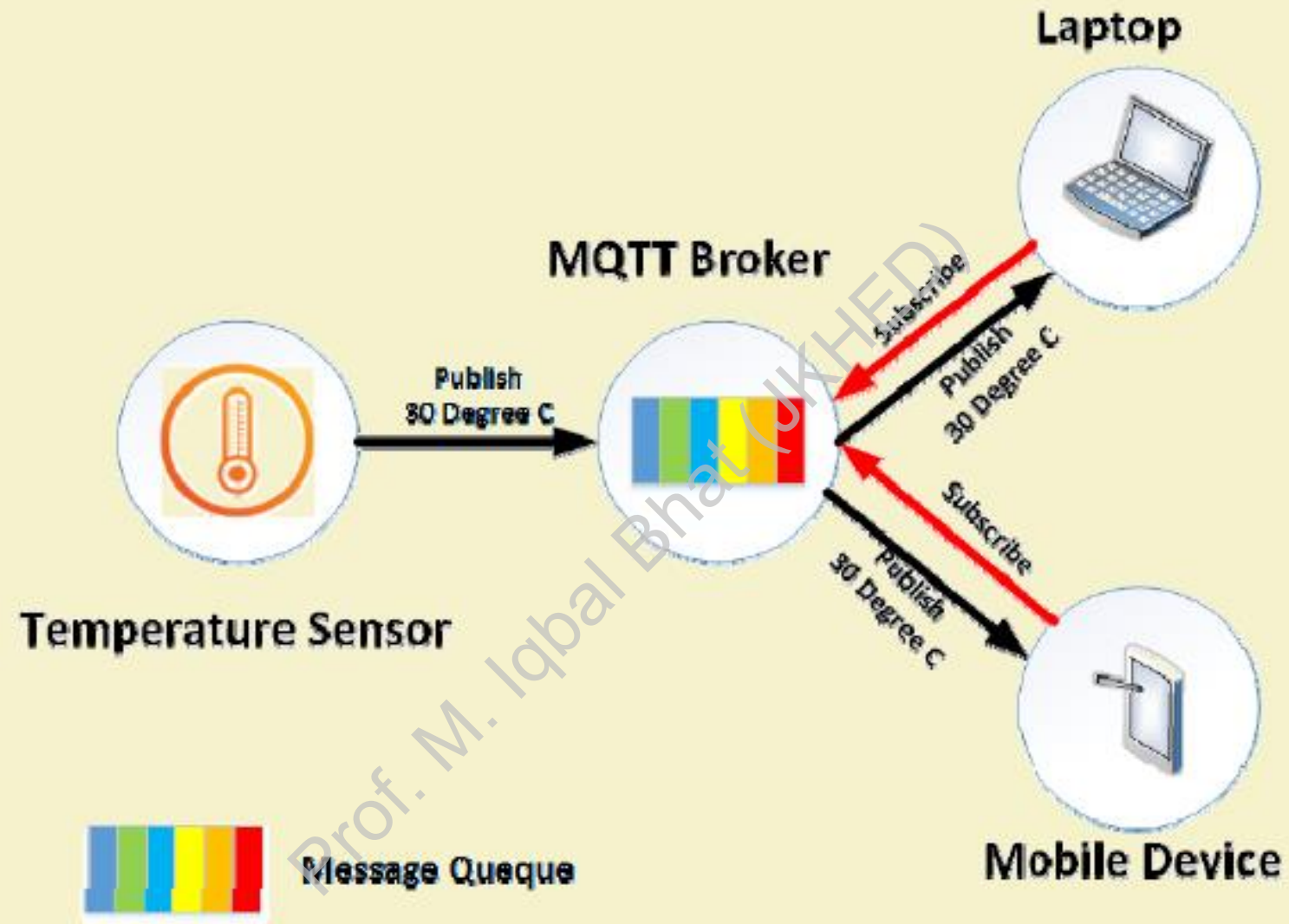
Disconnect

Subscribe

Unsubscribe

Publish

Source: ["MQTT", Wikipedia \(Online\)](#)



Communication

- ✓ The protocol uses a **publish/subscribe** architecture (HTTP uses a request/response paradigm).
- ✓ Publish/subscribe is **event-driven** and enables messages to be pushed to clients.
- ✓ The central **communication point is the MQTT broker**, which is in charge of dispatching all messages between the senders and the rightful receivers.
- ✓ Each client that publishes a message to the broker, includes a **topic** into the message. The **topic is the routing information for the broker**.

Source: [“MQTT 101 –How to Get Started with the lightweight IoT Protocol”, HiveMQ \(Online\)](#)

- ✓ Each client that wants to receive messages subscribes to a certain topic and the broker delivers all messages with the matching topic to the client.
- ✓ Therefore the clients don't have to know each other. They only communicate over the topic.
- ✓ This architecture enables highly scalable solutions without dependencies between the data producers and the data consumers.

Source: [“MQTT 101 –How to Get Started with the lightweight IoT Protocol”, HiveMQ \(Online\)](#)

MQTT Topics

- ✓ A topic is a **simple string** that can have more hierarchy levels, which are separated by a slash.
- ✓ A sample topic for sending temperature data of the living room could be *house/living-room/temperature*.
- ✓ On one hand the client (e.g. mobile device) can subscribe to the exact topic or on the other hand, it can use a **wildcard**.

Source: [“MQTT 101 –How to Get Started with the lightweight IoT Protocol”, HiveMQ \(Online\)](#)

- ✓ The subscription to *house/+/temperature* would result in all messages sent to the previously mentioned topic *house/living-room/temperature*, as well as any topic with an arbitrary value in the place of living room, such as *house/kitchen/temperature*.
- ✓ The plus sign is a **single level wild card** and only allows arbitrary values for one arbitrary level.
- ✓ If more than one level needs to be subscribed, such as, the entire sub-tree, there is also a **multilevel wildcard** (#).
- ✓ It allows to subscribe to all underlying hierarchy levels.
- ✓ For example *house/#* is subscribing to all topics beginning with *house*.

Source: [“MQTT 101 –How to Get Started with the lightweight IoT Protocol”, HiveMQ \(Online\)](#)

Applications

- ✓ **Facebook Messenger** uses MQTT for online chat.
- ✓ **Amazon Web Services** use Amazon IoT with MQTT.
- ✓ **Microsoft Azure** IoT Hub uses MQTT as its main protocol for telemetry messages.
- ✓ The **EVERYTHING IoT platform** uses MQTT as an M2M protocol for millions of connected products.
- ✓ **Adafruit** launched a free MQTT cloud service for IoT experimenters called Adafruit IO.

SMQTT

- ✓ **Secure MQTT** is an extension of MQTT which uses encryption based on lightweight attribute based encryption.
- ✓ The main advantage of using such encryption is the broadcast encryption feature, in which one message is encrypted and delivered to multiple other nodes, which is quite common in IoT applications.
- ✓ In general, the algorithm consists of four main stages: setup, encryption, publish and decryption.

Source: M. Singh, M. Rajan, V. Shivraj, and P. Balamuralidhar, "Secure MQTT for Internet of Things (IoT)," in Fifth International Conference on Communication Systems and Network Technologies (CSNT 2015), April 2015, pp. 746-751

- ✓ In the setup phase, the subscribers and publishers register themselves to the broker and get a master secret key according to their developer's choice of key generation algorithm.
- ✓ When the data is published, it is encrypted and published by the broker which sends it to the subscribers, which is finally decrypted at the subscriber end having the same master secret key.
- ✓ The key generation and encryption algorithms are not standardized.
- ✓ SMQTT is proposed only to enhance MQTT security features.

Source: M. Singh, M. Rajan, V. Shivraj, and P. Balamuralidhar, "Secure MQTT for Internet of Things (IoT)," in Fifth International Conference on Communication Systems and Network Technologies (CSNT 2015), April 2015, pp. 746-751

